"This code was clearly overdesigned," Bill admits. "We could have just defined a regular method that took the person's name as a string—but we'd just discovered method_missing, so we used *Ghost Methods (57)* instead. That wasn't a good idea; the code didn't work as expected."

Can you spot the problem with the Roulette class? If you can't, try running it on your computer. Now can you explain what is happening?

Quiz Solution

The Roulette contains a bug that causes an infinite loop. It prints a long list of numbers and finally crashes.

```
2...
7...
1...
5...
(...more numbers here...)
roulette_failure.rb:7:in `method_missing': stack level too deep (SystemStackError)
```

This bug is nasty and difficult to spot. The variable number is defined within a block (the block that gets passed to times) and falls out of scope by the last line of method_missing. When Ruby executes that line, it can't know that the number there is supposed to be a variable. As a default, it assumes that number must be a parentheses-less method call on self.

In normal circumstances, you would get an explicit NoMethodError that makes the problem obvious. But in this case you have a method_missing, and that's where the call to number ends. The same chain of events happens again—and again and again—until the call stack overflows.

This is a common problem with Ghost Methods: because unknown calls become calls to method_missing, your object might accept a call that's just plain wrong. Finding a bug like this one in a large program can be pretty painful.

To avoid this kind of trouble, take care not to introduce too many Ghost Methods. For example, Roulette might be better off if it simply accepted the names of people on Frank's team. Also, remember to fall back on BasicObject#method_missing when you get a call you don't know how to handle. Here's a better Roulette that still uses method_missing:

methods/roulette_solution.rb

```
class Roulette
  def method_missing(name, *args)
    person = name.to_s.capitalize
    super unless %w[Bob Frank Bill].include? person
    number = 0
```